

ABC INSTITUTE OF TECHNOLOGY
AUTONOMOUS EXAMINATIONS BRANCH, HYDERABAD
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
DAA FINAL EXAMINATION

Branch / Dept: COMPUTER SCIENCE AND ENGINEERING	Course Code: R20-CSE	Subject Code: CS601PC
Subject: Design and Analysis of Algorithms	Date of Exam: 15-11-2026	Max Marks: 30 Marks
Semester / Year: III Year II Semester	Academic Year: 2026-2027	Time / Duration: 3 Hours

General Instructions:

- 1. Answer all questions in Part A.
- 2. In Part B, answer either (a) or (b) from each question.
- 3. Assume suitable missing data if necessary.

PART A

DESCRIPTIVE SECTION

Instructions: Answer any 3 questions out of 5. [5 Questions × 10M | Attempt any 3 | Max Marks: 30]

1. Critically evaluate Greedy Method against alternative paradigms. When is Greedy Method strictly optimal? **[10 Marks]**
2. Critically evaluate Knapsack Problem against alternative paradigms. When is Knapsack Problem strictly optimal? **[10 Marks]**
3. Provide a comprehensive, in-depth analysis of Job Sequencing. Discuss its algorithmic or theoretical design in detail. **[10 Marks]**
4. Derive and explain the mathematical recurrence or governing equations of Optimal Merge Patterns. Prove its correctness. **[10 Marks]**
5. Provide a comprehensive, in-depth analysis of 0/1 Knapsack. Discuss its algorithmic or theoretical design in detail. **[10 Marks]**